AI Playbook and Toolkit for Public Health Departments

Equity Evaluation for AI Models

ANL 330

Equity evaluation examines whether an AI model performs differently across populations, places, languages, programs, or service contexts in ways that could create or worsen public health inequities. Public health agencies have a responsibility to evaluate not only average model performance but also who benefits, who may be missed, and who may be harmed. This is especially important when AI is used to prioritize cases, allocate services, detect risk, summarize records, or guide outreach. This module teaches learners how to evaluate AI models and AI-supported outputs through an equity lens. It emphasizes subgroup performance, missingness, representativeness, access, language, disability, rurality, socioeconomic context, and public health impact. The goal is to help agencies identify equity risks before deployment and monitor them after launch. Equity evaluation is not a separate ethical add-on. It is a technical and operational requirement for responsible public health AI. A model that performs well overall may still fail for groups that are underrepresented, incorrectly coded, less connected to healthcare systems, or affected by structural barriers.

Level	intermediate/practitioner
Primary track	Analytics and Modeling Track
Appears in tracks	analytics-and-modeling, governance-and-security, epidemiology, policy, communications

Learning Objectives

- Explain why overall model performance can mask inequitable performance.
- Identify relevant subgroups and contextual factors for equity evaluation in public health.
- Assess data representativeness, missingness, and measurement bias before model use.
- Define subgroup performance measures and review thresholds for AI-supported workflows.
- Produce an equity evaluation plan for a model or AI-supported workflow.

Module Overview

Equity evaluation examines whether an AI model performs differently across populations, places, languages, programs, or service contexts in ways that could create or worsen public health inequities. Public health agencies have a responsibility to evaluate not only average model performance but also who benefits, who may be missed, and who may be harmed. This is especially important when AI is used to prioritize cases, allocate services, detect risk, summarize records, or guide outreach.

This module teaches learners how to evaluate AI models and AI-supported outputs through an equity lens. It emphasizes subgroup performance, missingness, representativeness, access, language, disability, rurality, socioeconomic context, and public health impact. The goal is to help agencies identify equity risks before deployment and monitor them after launch.

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Jurisdiction and Agency Policy Note

This module provides national-level training guidance for public health agencies and partners. It is not legal advice and does not replace review by agency legal counsel, privacy officers, procurement officials, civil rights staff, records officers, or other authorized decision-makers. Requirements may vary by state, locality, territory, Tribe, agency, funding source, data-sharing agreement, emergency authority, and organizational policy. Learners should use this module to identify the issues that require review and should confirm applicable requirements before approving, procuring, deploying, or publicly communicating about AI-supported work.

Public Health Example

A model predicts which clients are most likely to miss follow-up care after a public health referral. Overall performance appears acceptable, but subgroup review shows lower sensitivity for rural clients and people with missing phone numbers. The model may under-prioritize people who face transportation, broadband, housing, or communication barriers. Equity evaluation would require additional analysis, workflow changes, and monitoring before the tool is used for outreach prioritization.

Technical and Operational Deep Dive

Equity evaluation begins with the intended use. A model used for internal workload planning has different equity implications than a model used to prioritize field response or determine which communities receive outreach. Evaluators should identify who is affected by the model output, what action follows, and what harms may occur from false positives, false negatives, or delayed review.

Data assessment is the next step. Public health datasets often reflect uneven healthcare access, reporting patterns, documentation practices, language access, and program reach. Missing race, ethnicity, language, disability, address, housing, or contact information may not be random. It may reflect structural barriers or data collection practices that affect some communities more than others.

Subgroup performance should be measured when data allow. This may include sensitivity, specificity, false positive rates, false negative rates, calibration, timeliness, or user override rates by race, ethnicity, age, sex, gender, geography, language, disability, rurality, payer, facility type, or other relevant factors. Subgroups should be chosen based on public health relevance, not simply data convenience.

Equity evaluation should lead to action. Mitigations may include changing thresholds, improving data collection, adding human review, limiting the intended use, creating separate workflows, improving language access, consulting community partners, adding uncertainty warnings, or declining to deploy the model. Equity monitoring should continue after launch because data patterns and workflow use can change.

Risks, Failure Modes, and Guardrails

Risks and failure modes include the following:

Relying on average performance while ignoring subgroup harms.

Using datasets that underrepresent communities with limited healthcare access.

Treating missing data as neutral when missingness reflects structural inequities.

Using AI prioritization to allocate scarce resources without equity review.

Failing to monitor equity impacts after deployment.

Recommended guardrails include the following:

Require equity evaluation for AI uses that affect prioritization, eligibility, outreach, response, or public communication.

Assess representativeness and missingness before model use.

Measure subgroup performance where feasible and document limitations where not feasible.

Include community, equity, and program reviewers in governance decisions.

Monitor equity indicators and user overrides after deployment.

Reflection Questions

Where does this topic appear in your agency, program, or workflow?

Which staff roles would need to understand or apply this concept before AI use is approved?

What could go wrong if this topic is not addressed before implementation?

What evidence would governance, leadership, or operations staff need before moving forward?

Which policy, data, equity, privacy, security, or workflow questions remain unresolved?

Practical Exercise

Create an equity evaluation plan for one AI model or AI-supported workflow. Include intended use, affected populations, relevant subgroups, data representativeness concerns, missingness concerns, subgroup performance measures, potential harms, mitigation options, monitoring indicators, and decision criteria.

Expected Artifact or Evidence

Equity evaluation plan

Subgroup and data representativeness review

Potential harm and mitigation table

Equity monitoring indicators

Governance recommendation

Expected Assignment

- Equity evaluation plan
- Subgroup and data representativeness review
- Potential harm and mitigation table
- Equity monitoring indicators
- Governance recommendation

Knowledge Check

- 1. Why is overall model performance insufficient for public health equity review?
- 2. What is measurement bias?
- 3. Which use most clearly requires equity evaluation?
- 4. What should happen if subgroup performance is poor for a population affected by the workflow?
- 5. What is strong completion evidence?

References and Resources

- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP guidance for LLM application security
- Agency policies for privacy, cybersecurity, records, procurement, accessibility, and public communications
- Relevant public health program SOPs, data governance documentation, and implementation plans